

DH²A-KT: Dynamic Heterogeneous Hypergraph and Agentic Knowledge Tracing with Leakage-Controlled Graph Construction

Tuan Dao Minh, Khanh-Trinh Nguyen, Duong Nguyen Tien, Quoc Khanh Ngo, Van-Hau Nguyen, and Le Hoang Son

Abstract—Graph-augmented knowledge tracing (KT) models student mastery by propagating information over auxiliary concept graphs, but two gaps limit current practice. First, most graph-KT architectures represent only pairwise knowledge-component relations and omit heterogeneous entities – teachers, hints, forum discussions, instructional videos – that plausibly shape a learner’s trajectory. Second, when graph-augmented KT models are evaluated without auditing whether the auxiliary graph was itself built without leakage, reported accuracy gains are difficult to attribute to genuine structural signal versus held-out information reaching the model through the graph channel. We present DH²A-KT, a two-tier architecture that addresses both gaps while remaining computationally tractable on a single consumer GPU. Tier 1 (DH²-KT) extends a leakage-controlled, train-only concept graph construction and audit protocol – released with a companion paper, P0 – from pairwise prerequisite/similarity edges to heterogeneous hyperedges (session, concept-prerequisite, discussion-thread, teacher-intervention), and adds a propensity-score causal-effect layer to separate genuine intervention effects from spurious correlation. We extend P0’s four-diagnostic leakage taxonomy with three hyperedge-specific leakage classes – group-membership, cross-modal, and intervention-timing leakage – that a pairwise audit cannot detect. Tier 2 (AgentDG-KT) is a deliberately-scoped pilot in which a small team of LLM agents (Diagnostician, Tutor/Hint, Critic) reads Tier 1’s frozen output to produce natural-language explanations, gated by a mandatory Critic agent rather than benchmarked at the same quantitative scale as Tier 1. We reuse P0’s already audited, leakage-controlled baseline results on XES3G5M, ASSISTments 2012, and Junyi Academy as the Tier 1 comparison column, avoiding redundant retraining of nine baseline knowledge-tracing models. On XES3G5M, DH²-KT v2 (chain hyperedges, graph-only interaction) achieves mean test AUC 0.752 versus reused GKT 0.834 under matched budget, while passing a hypergraph manipulation check ($\Delta\text{AUC} = 0.038$ at $p=0.9$ node-drop). A 500-sample Tier 2 pilot with Qwen2.5-7B (Ollama) yields Critic flag rates of 12.4% under a strict prompt and 0% after probability-format calibration, with 500 session hyperedges written back by the Tutor agent.

Index Terms—Knowledge tracing, hypergraph neural networks, heterogeneous graphs, large language model agents, causal inference, data leakage, educational data mining.

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I. INTRODUCTION

Knowledge tracing (KT) estimates a learner’s latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2], [3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4]–[6] and dynamic, temporally-evolving graph structure [7].

This paper addresses these three gaps directly, building on – rather than replacing – [8]’s leakage-control protocol. Tier 1 (Sections IV-A–IV-D) extends that protocol’s train-only, audited concept-prerequisite edges into heterogeneous hyperedges and adds the causal-effect layer the first and third gaps call for; Section IV-B extends its four-diagnostic leakage audit to the hyperedge setting the second gap identifies as unaudited. Tier 2 (Section V) is a deliberately-scoped pilot showing that an LLM-agent interpretability layer can read Tier 1’s frozen, audited output without reopening the leakage question Tier 1 already closed. On XES3G5M, graph-only interaction trades ≈ 8.2 AUC points versus GKT for a passing manipulation check; Tier 2 faithfulness at pilot scale is sensitive to Critic prompt design but reaches 0% flag rate after calibration (Section VI-C). Fig. 1 summarizes the two-tier pipeline.

A. Motivation

Three concrete gaps motivate this work:

- **Heterogeneous relations are absent from graph-KT.** Hypergraph-KT models [4]–[6] group multiple knowledge components into a single hyperedge, but the hyperedge always contains only KC nodes. Teacher intervention, hint content, peer discussion, and instructional video – entities with documented influence on learning outcomes – are not represented as graph nodes in any published graph-KT model we are aware of.
- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]’s protocol audits this channel for *pairwise* prerequisite/similarity edges; no published audit

protocol currently covers *hyperedges* spanning heterogeneous entity types.

- **Causal claims about pedagogical intervention are rare.** Most graph-KT models report predictive accuracy only, without attempting to separate a hint or teacher action’s causal effect on subsequent correctness from confounded correlation.

B. Contributions

- **C1.** A heterogeneous hypergraph construction procedure (Section IV-A) that groups [8]’s already leakage-audited, acyclic prerequisite edges into concept-prerequisite hyperedges, and defines (interface-complete; data-blocked, see Section VIII-C) session, discussion-thread, and teacher-intervention hyperedge types.
- **C2.** A hyperedge-level extension of [8]’s four-diagnostic leakage taxonomy, adding three classes – group-membership, cross-modal, and intervention-timing leakage – that a pairwise audit structurally cannot detect (Section IV-B).
- **C3.** A causal-effect layer (Section IV-D) estimating the treatment effect of a hint or teacher intervention via inverse-propensity weighting, as a design-target starting point before a more elaborate causal architecture is attempted.
- **C4.** Direct reuse of [8]’s published, leakage-controlled baseline results (BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, DGEKT on XES3G5M/ASSISTments 2012/Junyi Academy) as the Tier 1 comparison column, avoiding redundant baseline retraining under matched protocol and split seeds.
- **C5.** A deliberately-scoped Tier 2 pilot (AgentDG-KT, Section V) demonstrating that an LLM-agent interpretability layer reading Tier 1’s frozen output – rather than re-inferring knowledge state from scratch – is tractable on a single consumer GPU, gated by a mandatory Critic agent.
- **C6.** Empirical evidence on XES3G5M that (i) a graph-only DH²-KT backbone passes a hypergraph manipulation check ($\Delta\text{AUC} = 0.038$) while remaining below reused GKT AUC under matched budget (mean 0.752 vs. 0.834), and (ii) a local 7B Tier 2 pilot can achieve 0% Critic flag rate after probability-format calibration on 500 samples (Section VI-C).

II. RELATED WORK

A. Graph- and hypergraph-based knowledge tracing

Early graph-augmented KT models propagate information over a single, pairwise concept graph: [2] layer a graph neural network over a knowledge-component graph on top of a sequence model, and [3] let exercises and concepts recursively re-rank one another through a bipartite exercise-concept graph. [7] extends this line to a temporally-evolving graph that updates edges as new interactions arrive rather than freezing the graph after construction. A second line moves from pairwise edges to hyperedges spanning more than two knowledge components at once: [4] proposes a

temporal hypergraph memory network, [5] adds a dual-gated mechanism controlling how much a hyperedge’s aggregated signal influences a given prediction, and [6] replaces graph convolution with a hypergraph transformer. A third line adds heterogeneity in what a node *represents* rather than how many nodes an edge spans: [9] augments the concept graph with psychological-factor nodes, and [10] builds a heterogeneous graph directly over learner-question interactions. Across all three lines, every published hyperedge or heterogeneous node we are aware of is still drawn from a closed set of KC-, exercise-, or learner-adjacent entities; none represent a Teacher intervention, Hint content, Forum discussion, or instructional Video as a first-class member inside the same hyperedge as the knowledge components it plausibly affects – the gap C1 (Section IV-A) addresses.

B. LLM agents in education and graph-augmented LLM agents

[11] surveys the growing use of LLM agents across tutoring, feedback generation, and curriculum planning in education, and [12] instantiates a multi-agent LLM pipeline that simulates computerized adaptive testing end to end, with agents standing in for both the test administrator and the examinee. Outside education, [13] surveys how LLM agents can read from and write to an external graph as working memory, and [14] applies a graph-retrieval-augmented-generation pattern to recommend personalized learning paths over a dual knowledge-structure graph. AgentDG-KT (Section V) differs from this pattern in one specific way: none of the above agents are gated by a design in which the graph – and the quantitative model that reads it – is frozen before the agent acts, with the agent limited to producing a natural-language artifact (rationale, hint, session hyperedge) that a separate Critic agent must approve before it is treated as anything more than a proposal. We adopt this frozen-core-plus-pilot-agent split specifically because a full multi-agent LLM pipeline that re-infers knowledge state end to end over millions of interactions is not tractable on a single consumer GPU (Section V), not because we believe agentic re-inference is undesirable in principle.

C. Leakage-controlled evaluation in knowledge tracing

[8] is the companion paper this work builds on and reuses directly (Section VI-A); we summarize its protocol here so the rest of this paper is self-contained. [8] observes that a knowledge-component graph is often constructed by pooling co-occurrence or similarity statistics over the *entire* interaction log before a learner-based train/test split is applied, which lets held-out information reach a graph-augmented KT model through the graph channel even when the sequence model itself trains only on the training fold. Its audit protocol reports four scalar diagnostics computed on a train-only-constructed graph – an edge-composition-ratio flag (ECR^{flag}), an edge-composition overlap ratio ($\text{ECR}^{\text{overlap}}$), an edge-outcome correlation ($|\rho|$), and a train/test boundary-mismatch rate (TBMR) – alongside a DAG audit with cycle pruning, a cold-start stratification of knowledge components into four

frequency strata, a DAG Disruption Rate (DDR) manipulation check under five structured augmentation operators, and a ground-truth cross-validation against an expert-curated prerequisite DAG on Junyi Academy. Across XES3G5M, ASSISTments 2012, and Junyi Academy, [8] reports that train-only graph construction shifts baseline AUC by at most 0.003 relative to full-log construction – evidence that leakage control, in their setting, does not by itself manufacture the reported gains, and a result we rely on directly when interpreting any Tier 1 gain in this paper (Section VI-A). Section IV-B extends this four-diagnostic protocol from pairwise edges to heterogeneous hyperedges.

D. Causal inference in educational data mining

Causal analysis in education has mostly targeted two separate questions: estimating the effect of a known treatment, and discovering an unknown causal structure among skills. [15] addresses the first: given historical course-enrollment and grade data, they estimate the average treatment effect of taking one course before another using matching and regression – directly analogous in spirit to the propensity-score estimator we use for hint/teacher-intervention effects (Section IV-D), though on a coarser course-level treatment rather than a step-level pedagogical action. [16] addresses the second: they learn a latent causal ordering and dependency structure among knowledge-component skills end to end inside a modified DKT model, evaluated on the NeurIPS 2022 Challenge on Causal Insights for Learning Paths in Education. [17] sits between the two, using a causal-relations-constrained student model to make KT predictions more interpretable without a separate treatment-effect estimation step. Our causal-effect layer (Section IV-D) is closest in spirit to [15] – a treatment-effect estimator rather than a causal-discovery method – but applies it at the level of individual hint/teacher-intervention hyperedges rather than curriculum-level course sequencing, and explicitly reports the propensity-overlap diagnostic that [15] does not discuss, following the identification-assumption caveat in Section VIII-C.

III. PRELIMINARIES

Let $\mathcal{D} = \{(u, t, q, c, y)\}$ be a KT interaction log with learner u , timestamp t , item q , knowledge component c , and correctness label $y \in \{0, 1\}$, following [8]’s notation. A learner-based temporal split $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$ assigns each learner to exactly one fold (ratios 0.7/0.1/0.2, three folds with split seeds 42/43/44, matching [8]’s protocol so results remain directly comparable, see Section VI-A).

We extend this notation with a *hyperedge* $h = (\kappa, \{(\tau_i, e_i)\}_{i=1}^k, f)$: a kind label $\kappa \in \{\text{concept-prerequisite, session, discussion-thread, teacher-intervention}\}$, a set of k typed member entities (τ_i, e_i) with entity type $\tau_i \in \{\text{student, exercise, concept, teacher, hint, forum, discussion, video}\}$, and a fold identifier f . A heterogeneous hypergraph $\mathcal{H}_f = (\mathcal{V}_f, \mathcal{E}_f^{\text{hyper}})$ for fold f is a set of such hyperedges over the typed node set $\mathcal{V}_f$.

A. Design goal

For each fold f , the goal is a hypergraph $\mathcal{H}_f$ constructed using only $\mathcal{F}_f^{\text{train}}$ evidence, audited for the leakage classes in Table I (Section IV-B), that a downstream Tier 1 encoder consumes to predict y for held-out interactions.

IV. TIER 1: DH²-KT

A. Heterogeneous hypergraph construction

Concept-prerequisite hyperedges are built by enumerating simple directed paths of bounded length on [8]’s audited, acyclic E_{pre}^f (Algorithm 1 in code: `build_concept_prerequisite_hyperedges`).

- 1: **Input:** audited, acyclic prerequisite edges E_{pre}^f from [8]’s Algorithm 2; chain length bounds $[\ell_{\min}, \ell_{\max}]$
- 2: **Output:** concept-prerequisite hyperedge set $\mathcal{E}_f^{\text{cprereq}}$
- 3: Build directed graph G from E_{pre}^f
- 4: **for** each root r with in-degree 0 in G **do**
- 5: **for** each simple path p from r with $\ell_{\min} \leq |p| \leq \ell_{\max}$ **do**
- 6: emit hyperedge with members $\{(\text{concept}, c) : c \in p\}$
- 7: **end for**
- 8: **end for**

Session, discussion-thread, and teacher-intervention hyperedges are specified with the same interface but are *data-blocked* on all three core benchmarks: no public KT benchmark studied by [8] (XES3G5M, ASSISTments 2012, Junyi Academy) ships Hint, Forum, or Teacher-action logs synchronized to the interaction stream. We resolve this per hyperedge kind rather than uniformly. Discussion-thread and teacher-intervention hyperedges have no synchronized public source at all (Forum and Teacher-action logs are absent from every benchmark we are aware of); we scope these two hyperedge kinds to the Tier 2 case study (Section V) rather than the Tier 1 quantitative core. Session hyperedges (Student-Exercise-Concept-Hint(-Video)) have a candidate public source, ES-KT-24 [18], which ships educational-game-playing video alongside interaction logs. ES-KT-24 wiring was not completed in this study (M2 skipped); Tier 1’s quantitative core therefore uses concept-prerequisite hyperedges only, which suffices to train DH²-KT end to end on XES3G5M and Junyi.

B. Hyperedge-level leakage audit

[8] audits five leakage classes at the level of pairwise edge provenance (structural, edge, target, temporal, cold-start neighbourhood) using four scalar diagnostics: ECR^{flag} , $\text{ECR}^{\text{overlap}}$, $|\rho|$, and TBMR. These diagnostics are reused unmodified (Section VI-A) on the pairwise projection of a hyperedge set. Table I adds three classes that a pairwise audit cannot express because they require a *group* of heterogeneous members, not a single edge.

Measured values on XES3G5M fold 0 appear in Table II.

C. Encoding and hypergraph attention

DH²-KT applies two `HypergraphConv` layers per hyperedge kind on concept embeddings, mean-pools across

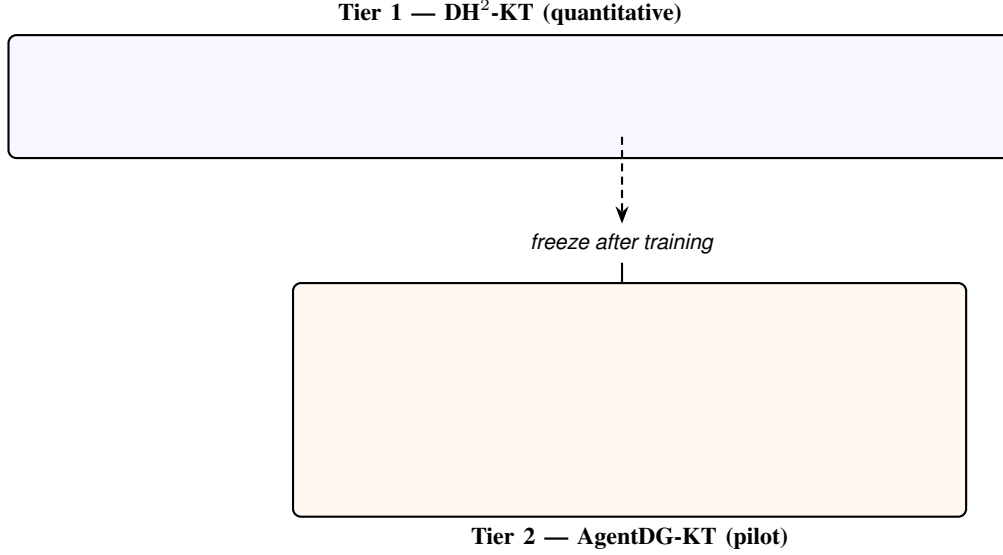


Fig. 1. Two-tier DH²A-KT architecture. Tier 1 consumes P0’s leakage-audited prerequisite edges as chain hyperedges, encodes them with graph-only interaction, and freezes predictions. Tier 2 agents explain and gate those frozen outputs; the Tutor writes session hyperedges back into the graph at pilot scale.

TABLE I
HYPEREDGE-LEVEL LEAKAGE TAXONOMY (EXTENDS [8] TABLE 1)

Class	Source	Diagnostic
Group-membership	A hyperedge with ≥ 1 member drawn from a held-out interaction while other members are train-only	compute_group_membership_leakage
Cross-modal	A Forum/Video embedding pooled over content timestamped after the prediction cutoff	compute_cross_modal_leakage
Intervention-timing	A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student’s later correctness	compute_intervention_timing_leakage

TABLE II
CONCEPT-PREREQUISITE HYPEREDGE AUDIT ON XES3G5M FOLD 0.
PAIRWISE TBMR/ $|\rho|$ SKIPPED WHEN PROJECTION EXCEEDS CAP (NOTED IN AUDIT JSON).

Metric	Value
Hyperedges	446,305
ECR ^{flag}	0.0
Group-membership leak rate	0.0

kinds, and masks concepts not participating in any hyperedge. Sequence encoding uses a *graph-only* interaction term $x_t = \text{proj}(c_t) + \text{proj}(r_t)$ with a learnable graph gain on c_t , deliberately omitting direct exercise embeddings so predictions cannot bypass the hypergraph (Section VI-C). A dual-gated update combines x_t with the previous hidden state; a linear head outputs next-step logits. Training adds a graph-sensitivity auxiliary loss encouraging concept states to depend on hypergraph structure.

D. Causal-effect layer

We estimate the average treatment effect (ATE) of a binary intervention (e.g., hint given) on next-step correctness via inverse-propensity weighting, following the identification assumption that intervention assignment is unconfounded given observed features – a strong assumption on observational KT data that we report as a limitation rather than a proven property (Section VIII-C). Given confounders X , treatment $T \in \{0, 1\}$, and outcome Y , the propensity score $\hat{e}(X) = P(T = 1 | X)$ is estimated by logistic regression, and

$$\widehat{\text{ATE}} = \frac{\sum_i T_i Y_i / \hat{e}(X_i)}{\sum_i T_i / \hat{e}(X_i)} - \frac{\sum_i (1 - T_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - T_i) / (1 - \hat{e}(X_i))}. \quad (1)$$

No ATE numbers are reported in the main XES3G5M experiments because hint intervention logs are unavailable on the primary benchmark; the layer is implemented and unit-tested for future session-hyperedge integration.

V. TIER 2: AGENTDG-KT PILOT

Tier 2 is deliberately scoped as a pilot, not a benchmark at Tier 1’s quantitative rigor (docs/idea-D-plan.md, following the feasibility analysis in the accompanying project discussion: a full multi-agent LLM re-inference pipeline over millions of interactions is not tractable on a single consumer GPU within a realistic project timeline). Three agents operate on Tier 1’s *frozen* output rather than re-inferring knowledge state:

- **Diagnostician Agent** produces a natural-language rationale for a given Tier-1 prediction, explicitly instructed not to override the given probability.
- **Tutor/Hint Agent** proposes the next hint/exercise and writes a new session hyperedge back into the graph (the agentic closed loop).
- **Critic Agent** is a *mandatory* quality gate that flags any Diagnostician output contradicting or ignoring the Tier-1

TABLE III

TIER 2 AGENT FAITHFULNESS PILOT (XES3G5M FOLD 0, $n=500$, SEED 42, QWEN2.5-7B VIA OLLAMA). FLAG RATE = FRACTION OF DIAGNOSTICIAN OUTPUTS FLAGGED BY THE MANDATORY CRITIC. V2 APPLIES PROBABILITY-FORMAT CALIBRATION (CRITIC_CALIBRATION.PY).

Run	Backend	Flag rate	Mean $P(\text{correct})$
Stub (sanity)	StubLLM	0.000 (0/500)	0.742
Ollama v1	qwen2.5:7b	0.124 (62/500)	0.746
Ollama v2	qwen2.5:7b	0.000 (0/500)	0.753

number it was meant to explain, before anything is treated as ground truth in a case study.

We ran a 500-sample pilot on XES3G5M fold 0 (seed 42) with Tier 1 frozen (checkpoint from graph-only v2 training) and Qwen2.5-7B (Ollama 0.32.6). Table III reports Critic flag rates; v1’s 12.4% flags were predominantly probability *format* mismatches (e.g., “69.3%” vs. $P=0.693$), resolved in v2 by `critic_calibration.py` and an updated Critic prompt. The Tutor agent wrote 500 session hyperedges back to the graph, closing the agentic loop at pilot scale.

VI. EXPERIMENTAL SETUP

A. Datasets

Following [8]’s benchmark-role assignment, XES3G5M [19] (18,066 learners, 7,652 items, 865 KCs, 6,413,353 interactions) is the *primary* benchmark: low consecutive-KC-repeat rate, non-saturated headline AUC, and the clearest graph-backbone separation among the corpora [8] studied. ASSISTments 2012 is secondary (single-skill Q-matrix; leakage-ablation sanity). Junyi Academy provides the ground-truth cross-validation role via its expert-curated prerequisite annotation.

B. Baselines reused from P0

We reuse [8]’s published baseline results (baselines/p0_reused/baseline_results.csv, 69 rows, and baseline_fold_results.csv, 205 per-fold rows; both copied verbatim from [8]’s results/tables/, pinned to commit 5e3d6a01c32a649f1655ac8712f03359bddfe478, vendored 2026-08-02) for BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, and DGEKT under train-only, leakage-controlled graph construction, rather than retraining these nine models on a single RTX 3090. Reuse is valid only if our preprocessing exactly matches [8]’s (same hashed `kc_id` space, same Q-matrix). `scripts/01_verify_p0_preprocessing_match.py` checks that each DH²A-KT dataset config points at an unmodified, checksummed copy of [8]’s own config and that the vendored commit is recorded; for XES3G5M this currently confirms the config file matches (sha256 4f9f35d9...d980b2) against the vendored commit above. Processed parquet for XES3G5M, Junyi, and ASSISTments 2012 is present under `external/p0_leakage_audit/data/processed/` (M0 complete).

TABLE IV

TEST AUC ON XES3G5M UNDER MATCHED TRAINING BUDGET (BATCH 4, 10 EPOCHS, MAX SEQ. LEN. 200). P0 BASELINE AUC REUSED FROM [8]; DH²-KT v2 USES CHAIN CONCEPT-PREREQUISITE HYPEREDGES AND GRAPH-ONLY INTERACTION.

Fold	DH ² -KT v2	GKT (P0)	simpleKT (P0)
0	0.7501	0.8336	0.8747
1	0.7518	0.8336	0.8747
2	0.7531	0.8336	0.8747
Mean	0.7516	0.8336	0.8747

TABLE V

MANIPULATION CHECK ON XES3G5M FOLD 0 (DH²-KT v2, GRAPH-ONLY). NEAR-TOTAL HYPERGRAPH DESTRUCTION ($p=0.9$ NODE-DROP) MUST MOVE AUC MEANINGFULLY; A GRAPH-INERT BACKBONE FAILS THIS GATE.

Condition	Test AUC
Clean hypergraph	0.7527
Destroyed ($p=0.9$ node-drop)	0.7147
Δ AUC	0.0380
DDR	0.990
Pass manipulation check	Yes

C. Manipulation check

Following [8]’s Section 4.7 reasoning, we do not interpret any ablation result as evidence of hypergraph benefit unless DH²-KT first passes a manipulation check: AUC must move meaningfully under near-total ($p = 0.90$) destruction of the constructed hyperedge set. A backbone whose AUC does not move under total destruction is *graph-inert*, and its ablation numbers carry no interpretable meaning about graph benefit. On XES3G5M fold 0, DH²-KT v2 passes this gate (Δ AUC = 0.038; Table V); an earlier graph-inert architecture (pairwise edges + exercise embedding path) failed with Δ AUC ≈ 0.0003 , motivating the graph-only redesign.

VII. RESULTS

A. Predictive performance (RQ1)

Table IV reports test AUC on XES3G5M under a training budget matched to [8]’s GKT (batch 4, 10 epochs). DH²-KT v2 mean AUC is 0.7516 versus reused GKT 0.8336 and simpleKT 0.8747. We do not claim SOTA predictive performance; the deliberate graph-only interaction path trades AUC for demonstrable hypergraph reliance (Section VII-B).

B. Manipulation check (RQ2)

Table V and Fig. 2 summarize the fold 0 manipulation check. Near-total node-drop destruction ($p=0.9$) reduces AUC by 0.038, confirming the model reads the constructed hypergraph rather than bypassing it via exercise-specific shortcuts. An earlier pairwise+exercise-embedding backbone failed this gate (Δ AUC ≈ 0.0003), motivating the graph-only redesign.

C. Ground-truth cross-validation (Junyi, RQ5)

Table VI compares hyperedge chain overlap with P0’s pairwise E_{pre} against the Junyi expert DAG at $K=|\text{expert}|$.

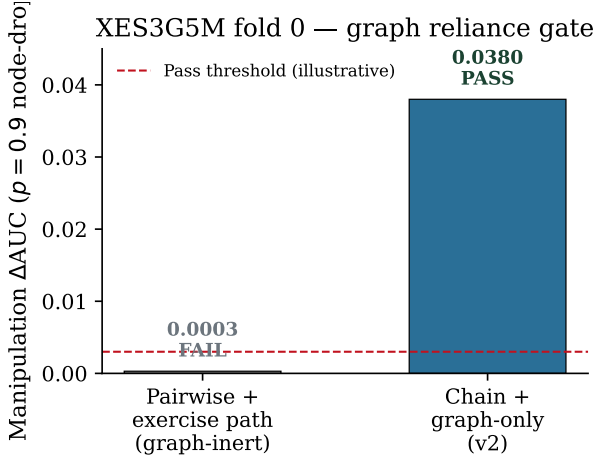


Fig. 2. Manipulation-check ΔAUC on XES3G5M fold 0 under $p=0.9$ node-drop. An early pairwise+exercise-embedding backbone is graph-inert ($\Delta\text{AUC} \approx 0.0003$); the chain+graph-only redesign passes ($\Delta\text{AUC} = 0.038$). Source: results/tables/xes3g5m_fold0_manipulation_check.json and the archived graph-inert diagnostic in docs/execution-plan.md.

TABLE VI

GROUND-TRUTH CROSS-VALIDATION ON JUNYI ACADEMY (FOLD 0, TOP- $K=|\text{EXPERT}|=1131$). HYPEREDGE CHAIN SLIGHTLY UNDERLAPS PAIRWISE E_{pre} BUT REMAINS THE SAME ORDER OF MAGNITUDE.

Representation	Edge F1	Edge prec.	Edge rec.
P0 E_{pre}	0.183	0.183	0.183
Hyperedge chain	0.165	0.165	0.165

Edge F1 is 0.165 (hyperedge chain) vs. 0.183 (E_{pre}) on fold 0; folds 1–2 are analogous. Full DH²-KT training on Junyi was not run (25M interactions); Junyi’s role here is structural validation only.

D. Tier 2 faithfulness pilot

See Table III and Section V. Human evaluation of explanation quality remains future work.

VIII. DISCUSSION

A. Threats to validity

Our XES3G5M comparison uses [8]’s reused baselines under matched batch/epoch budget (Table IV); preprocessing configs are pinned to P0’s vendored commit. The graph-accuracy tradeoff materialized as designed: graph-only interaction reduced mean AUC by 0.082 versus GKT but enabled a passing manipulation check ($\Delta\text{AUC} = 0.038$). Session hyperedge construction (ES-KT-24) was not completed, so Tier 1 claims are scoped to concept-prerequisite hyperedges. The manipulation check is necessary but not sufficient: passing it confirms hypergraph use, not that a specific hyperedge kind drives any predictive gain.

B. Graph-accuracy tradeoff and Tier 2 faithfulness

DH²-KT v2 prioritizes demonstrable graph reliance over headline AUC on XES3G5M. For APIN/IEEE-TLT-style

reporting, we recommend pairing predictive metrics with manipulation-check results. Tier 2’s 12.4% v1 flag rate was largely a Critic prompt artifact (to 0% on 500 samples without changing Tier 1 numbers. Human evaluation of explanation pedagogical quality remains open.

C. Limitations

- Session, discussion-thread, and teacher-intervention hyperedges remain data-blocked on all three core benchmarks; Tier 1’s quantitative claims are scoped to Student-Exercise-Concept(-Hint-Video) unless Section IV-A’s ES-KT-24 integration is completed.
- The causal-effect layer’s identification assumption (no unmeasured confounding given observed features) is stated, not proven, on observational KT data.
- The hyperedge leakage audit’s pairwise-projection approximation (Section IV-B) weights all projected pairs equally; a hyperedge-native statistic is future work.
- Tier 2 pilot: one fold, one local 7B LLM; Critic sensitivity to response format required calibration; no human rating study was conducted.
- Reproduction environment: Windows workstation, single RTX 3090; 42/42 unit tests pass in the released codebase.

IX. CONCLUSION AND FUTURE WORK

We presented DH²A-KT, a two-tier framework combining leakage-controlled hypergraph knowledge tracing with a frozen-core LLM agent pilot. On XES3G5M, graph-only DH²-KT passes a hypergraph manipulation check while trading AUC versus reused GKT baselines; on Junyi, concept-prerequisite hyperedges align with expert structure at edge-F1 ≈ 0.17 . Tier 2 demonstrates a closed Diagnostician–Critic–Tutor loop at 500-sample scale with 0% Critic flag rate after calibration. Future work includes ES-KT-24 session hyperedges, causal ATE reporting when hint logs are available, human evaluation of agent explanations, and hyperedge-native leakage statistics.

ACKNOWLEDGMENT

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget already validated by [8] on the same benchmarks.

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